

# Four-Dimensional Parameter Expansion Study

Generated by opt\_4dim.py

2026-05-06 18:31

## Purpose

This report extends the 3D side study from  $(k, \delta, x)$  to  $(k, \delta, x, y)$ , testing all selected pairs  $(x, y)$  from the remaining Layer 2 parameter set

$$\{q, \alpha, \beta, L, a, z_{\text{trend}}, \rho\}.$$

Pairs involving  $\rho$  use online smoothing inside each fold, because  $\rho$  does not affect the static single-fold strategy directly.

## Dataset, Protocol, and Search Grid

### Dataset and Protocol

|               |                     |
|---------------|---------------------|
| Data source   | Yahoo adjusted SPY  |
| Symbol        | SPY                 |
| Lookback      | 5 years             |
| Bar interval  | 1d                  |
| Bars          | 1255                |
| Train size    | 252 bars            |
| Test size     | 63 bars             |
| Cost per side | 0.0 bps             |
| Objective     | $R^m + D^m + 0.15G$ |

### Search Grid

| Parameter | Values searched                 |
|-----------|---------------------------------|
| k         | 0.8, 1.1, 1.4, 1.8, 2.2         |
| delta     | 1, 2, 3, 5, 8                   |
| q         | 0.0, 5e-08, 1e-07, 2e-07, 5e-07 |
| alpha     | 0.03, 0.06, 0.1, 0.15           |
| beta      | 0.8, 0.88, 0.9, 0.94            |
| L         | 30, 60, 90, 120                 |
| a         | 0.02, 0.04, 0.06, 0.08          |
| z_trend   | 0.0, 0.25, 0.35, 0.5, 0.75      |
| rho       | 0.1, 0.25, 0.4, 0.6             |

**Grid note.** Each 4D row searches  $(k, \delta, x, y)$ . This report uses the same configured search grid as the 3D report.

**Initialization and API data note.** Unless a row explicitly optimizes a parameter, the GUI-style continuous runs start from  $(k_0, \delta_0) = (1.2, 3)$  with  $q = 10^{-7}$ ,  $\alpha = 0.06$ ,  $\beta = 0.90$ ,  $L = 60$ ,  $a = 0.04$ ,  $z_{\text{trend}} = 0.35$ , and  $\rho = 0.25$ . The Yahoo API dataset is a rolling market-data snapshot fetched as of May 6, 2026, so rerunning the report on a later date may shift the bar count and the reported returns.

## Stitched Walk-Forward Results

| Experiment    | Optimized                  | MtM    | BH     | Gap     | Max DD  | Sharpe | Calmar | Trades | Avg hold | $\Delta$ MtM | Final selected                               |
|---------------|----------------------------|--------|--------|---------|---------|--------|--------|--------|----------|--------------|----------------------------------------------|
| baseline      | k, delta                   | 69.25% | 77.29% | -4.54%  | -18.76% | 0.90   | 0.80   | 5      | 33.20    | 0.00%        | k=1.2, delta=3                               |
| q+alpha       | k, delta, q, alpha         | 63.22% | 77.29% | -7.93%  | -18.76% | 0.85   | 0.74   | 5      | 32.20    | -6.02%       | k=0.8, delta=1, q=0, alpha=0.03              |
| q+beta        | k, delta, q, beta          | 61.34% | 77.29% | -9.00%  | -18.76% | 0.83   | 0.73   | 5      | 32.20    | -7.91%       | k=0.8, delta=1, q=0, beta=0.8                |
| q+L           | k, delta, q, L             | 60.38% | 77.29% | -9.54%  | -18.76% | 0.83   | 0.72   | 6      | 32.17    | -8.87%       | k=0.8, delta=1, q=5.0e-08, L=30              |
| q+a           | k, delta, q, a             | 67.73% | 77.29% | -5.39%  | -18.76% | 0.93   | 0.79   | 10     | 27.40    | -1.52%       | k=0.8, delta=1, q=5.0e-08, a=0.02            |
| q+z_trend     | k, delta, q, z_trend       | 63.22% | 77.29% | -7.93%  | -18.76% | 0.85   | 0.74   | 5      | 32.20    | -6.02%       | k=1.2, delta=3, q=0, z_trend=0               |
| q+rho         | k_init, delta_init, q, rho | 70.07% | 77.29% | -4.07%  | -18.76% | 0.95   | 0.81   | 9      | 26.33    | 0.82%        | k_init=0.8, delta_init=3, q=5.0e-08, rho=0.1 |
| alpha+beta    | k, delta, alpha, beta      | 65.62% | 77.29% | -6.58%  | -18.76% | 0.87   | 0.77   | 5      | 32.20    | -3.63%       | k=0.8, delta=3, alpha=0.03, beta=0.8         |
| alpha+L       | k, delta, alpha, L         | 64.63% | 77.29% | -7.14%  | -18.76% | 0.87   | 0.76   | 6      | 32.17    | -4.61%       | k=0.8, delta=1, alpha=0.03, L=30             |
| alpha+a       | k, delta, alpha, a         | 59.17% | 77.29% | -10.22% | -18.76% | 0.82   | 0.70   | 9      | 29.44    | -10.07%      | k=1.2, delta=3, alpha=0.03, a=0.02           |
| alpha+z_trend | k, delta, alpha, z_trend   | 66.64% | 77.29% | -6.00%  | -18.76% | 0.88   | 0.78   | 6      | 27.17    | -2.60%       | k=1.2, delta=3, alpha=0.06, z_trend=0        |

|              |                                  |        |        |         |         |      |      |    |       |         |                                               |
|--------------|----------------------------------|--------|--------|---------|---------|------|------|----|-------|---------|-----------------------------------------------|
| alpha+rho    | k_init, delta_init, alpha, rho   | 70.07% | 77.29% | -4.07%  | -18.76% | 0.95 | 0.81 | 9  | 26.33 | 0.82%   | k_init=0.8, delta_init=3, alpha=0.06, rho=0.1 |
| beta+L       | k, delta, beta, L                | 64.54% | 77.29% | -7.19%  | -18.76% | 0.87 | 0.76 | 6  | 32.17 | -4.71%  | k=0.8, delta=1, beta=0.8, L=30                |
| beta+a       | k, delta, beta, a                | 58.77% | 77.29% | -10.45% | -18.76% | 0.82 | 0.70 | 9  | 29.44 | -10.48% | k=1.2, delta=3, beta=0.8, a=0.02              |
| beta+z_trend | k, delta, beta, z_trend          | 66.64% | 77.29% | -6.00%  | -18.76% | 0.88 | 0.78 | 6  | 27.17 | -2.60%  | k=1.2, delta=3, beta=0.8, z_trend=0           |
| beta+rho     | k_init, delta_init, beta, rho    | 70.07% | 77.29% | -4.07%  | -18.76% | 0.95 | 0.81 | 9  | 26.33 | 0.82%   | k_init=0.8, delta_init=3, beta=0.8, rho=0.1   |
| L+a          | k, delta, L, a                   | 59.17% | 77.29% | -10.22% | -18.76% | 0.82 | 0.70 | 9  | 29.44 | -10.07% | k=1.2, delta=1, L=60, a=0.02                  |
| L+z_trend    | k, delta, L, z_trend             | 69.32% | 77.29% | -4.50%  | -18.76% | 0.90 | 0.80 | 7  | 28.57 | 0.07%   | k=0.8, delta=1, L=30, z_trend=0               |
| L+rho        | k_init, delta_init, L, rho       | 55.09% | 77.29% | -12.52% | -18.76% | 0.79 | 0.66 | 11 | 29.55 | -14.15% | k_init=0.8, delta_init=1, L=30, rho=0.1       |
| a+z_trend    | k, delta, a, z_trend             | 60.07% | 77.29% | -9.71%  | -18.76% | 0.83 | 0.71 | 11 | 25.27 | -9.17%  | k=1.2, delta=1, a=0.02, z_trend=0             |
| a+rho        | k_init, delta_init, a, rho       | 60.24% | 77.29% | -9.62%  | -18.76% | 0.86 | 0.71 | 13 | 29.15 | -9.01%  | k_init=0.8, delta_init=3, a=0.02, rho=0.25    |
| z_trend+rho  | k_init, delta_init, z_trend, rho | 70.67% | 77.29% | -3.73%  | -18.76% | 0.94 | 0.82 | 9  | 25.44 | 1.42%   | k_init=0.8, delta_init=3, z_trend=0, rho=0.1  |

## GUI-Style Continuous Online Results

| Experiment    | Swept              | Real.   | MtM     | BH     | Gap     | Max DD  | Sharpe | Calmar | Trades | Avg hold | $\Delta$ MtM | Selected + end live                          |
|---------------|--------------------|---------|---------|--------|---------|---------|--------|--------|--------|----------|--------------|----------------------------------------------|
| baseline      | fixed GUI defaults | 42.05%  | 53.40%  | 86.13% | -17.59% | -25.56% | 0.61   | 0.35   | 8      | 122.88   | 0.00%        | fixed GUI defaults, end k=0.8, end d=2       |
| q+alpha       | q, alpha           | 101.55% | 122.43% | 86.13% | 19.50%  | -17.68% | 1.09   | 0.98   | 16     | 57.19    | 69.03%       | q=0, alpha=0.03, end k=0.8, end d=2          |
| q+beta        | q, beta            | 86.48%  | 105.80% | 86.13% | 10.57%  | -17.68% | 0.98   | 0.88   | 15     | 65.33    | 52.40%       | q=0, beta=0.8, end k=0.8, end d=2            |
| q+L           | q, L               | 80.98%  | 99.73%  | 86.13% | 7.30%   | -17.68% | 0.94   | 0.84   | 14     | 72.07    | 46.33%       | q=0, L=60, end k=0.8, end d=2                |
| q+a           | q, a               | 80.98%  | 99.73%  | 86.13% | 7.30%   | -17.68% | 0.94   | 0.84   | 14     | 72.07    | 46.33%       | q=0, a=0.04, end k=0.8, end d=2              |
| q+z_trend     | q, z_trend         | 70.30%  | 87.94%  | 86.13% | 0.97%   | -18.76% | 0.84   | 0.72   | 14     | 77.21    | 34.54%       | q=0, z_trend=0.25, end k=0.8, end d=2        |
| q+rho         | q, rho             | 94.84%  | 115.03% | 86.13% | 15.52%  | -19.50% | 1.05   | 0.85   | 14     | 73.64    | 61.63%       | q=5.0e-08, rho=0.1, end k=0.8, end d=3       |
| alpha+beta    | alpha, beta        | 49.87%  | 65.40%  | 86.13% | -11.14% | -25.56% | 0.70   | 0.42   | 9      | 119.22   | 12.00%       | alpha=0.03, beta=0.88, end k=0.8, end d=2    |
| alpha+L       | alpha, L           | 49.87%  | 65.40%  | 86.13% | -11.14% | -25.56% | 0.70   | 0.42   | 9      | 119.22   | 12.00%       | alpha=0.03, L=60, end k=0.8, end d=2         |
| alpha+a       | alpha, a           | 77.09%  | 89.88%  | 86.13% | 2.01%   | -22.97% | 0.88   | 0.60   | 20     | 49.75    | 36.48%       | alpha=0.06, a=0.02, end k=0.8, end d=2       |
| alpha+z_trend | alpha, z_trend     | 54.59%  | 70.61%  | 86.13% | -8.34%  | -24.10% | 0.74   | 0.47   | 9      | 119.78   | 17.21%       | alpha=0.03, z_trend=0.25, end k=0.8, end d=2 |
| alpha+rho     | alpha, rho         | 99.72%  | 115.68% | 86.13% | 15.88%  | -19.50% | 1.09   | 0.86   | 15     | 61.93    | 62.28%       | alpha=0.06, rho=0.1, end k=0.8, end d=3      |
| beta+L        | beta, L            | 49.87%  | 65.40%  | 86.13% | -11.14% | -25.56% | 0.70   | 0.42   | 9      | 119.22   | 12.00%       | beta=0.8, L=60, end k=0.8, end d=2           |
| beta+a        | beta, a            | 70.85%  | 82.35%  | 86.13% | -2.03%  | -22.97% | 0.91   | 0.56   | 19     | 49.53    | 28.95%       | beta=0.8, a=0.02, end k=0.8, end d=2         |
| beta+z_trend  | beta, z_trend      | 54.59%  | 70.61%  | 86.13% | -8.34%  | -24.10% | 0.74   | 0.47   | 9      | 119.78   | 17.21%       | beta=0.8, z_trend=0.25, end k=0.8, end d=2   |
| beta+rho      | beta, rho          | 99.72%  | 115.52% | 86.13% | 15.79%  | -19.50% | 1.09   | 0.85   | 15     | 61.93    | 62.12%       | beta=0.88, rho=0.1, end k=0.8, end d=3       |
| L+a           | L, a               | 77.09%  | 89.88%  | 86.13% | 2.01%   | -22.97% | 0.88   | 0.60   | 20     | 49.75    | 36.48%       | L=60, a=0.02, end k=0.8, end d=2             |
| L+z_trend     | L, z_trend         | 74.34%  | 78.02%  | 86.13% | -4.36%  | -24.64% | 0.77   | 0.50   | 10     | 119.30   | 24.62%       | L=30, z_trend=0, end k=0.8, end d=2          |
| L+rho         | L, rho             | 99.72%  | 115.68% | 86.13% | 15.88%  | -19.50% | 1.09   | 0.86   | 15     | 61.93    | 62.28%       | L=60, rho=0.1, end k=0.8, end d=3            |
| a+z_trend     | a, z_trend         | 83.46%  | 96.70%  | 86.13% | 5.68%   | -22.97% | 0.93   | 0.63   | 21     | 48.71    | 43.30%       | a=0.02, z_trend=0.25, end k=0.8, end d=2     |
| a+rho         | a, rho             | 99.72%  | 115.68% | 86.13% | 15.88%  | -19.50% | 1.09   | 0.86   | 15     | 61.93    | 62.28%       | a=0.04, rho=0.1, end k=0.8, end d=3          |
| z_trend+rho   | z_trend, rho       | 102.48% | 118.08% | 86.13% | 17.16%  | -19.50% | 1.10   | 0.87   | 15     | 62.20    | 64.68%       | z_trend=0.25, rho=0.1, end k=0.8, end d=3    |

## High-Level Takeaways

- Stitched best MtM return: z\_trend+rho with 70.67%.
- Stitched best benchmark gap: z\_trend+rho with -3.73%.

- Stitched best Sharpe ratio:  $q + \rho$  with 0.95.
- GUI-style best MtM return:  $q + \alpha$  with 122.43%.
- GUI-style best benchmark gap:  $q + \alpha$  with 19.50%.
- GUI-style best Sharpe ratio:  $z_{\text{trend}} + \rho$  with 1.10.

## Caveats

The 4D grid is substantially larger than the 3D grid, so overfitting risk is higher. Treat this as a sensitivity and screening report, not final evidence. Pairs involving  $\rho$  are not perfectly comparable to static pairs because they use the online adaptation engine inside folds.